Ericsson Hebbal Edge PE Node HEB-PE-12 BGP/OSPF/ISIS Routing Protocol Flap SOP

Node: HEB-PE-12 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_9606

Description

This Standard Operating Procedure (SOP) addresses the occurrence of a Routing Protocol Flap failure on the Ericsson PE node HEB-PE-12 at the Hebbal Edge site. The failure domain is characterized by a series of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding events, ultimately resulting in the issuance of Alarm Code ALARM_9606 with a severity rating of Minor. This condition arises due to the intricate interplay between the routing protocols (BGP, OSPF, and ISIS) running on the PE node, which are responsible for discovering and maintaining routing information with neighboring routers. When an anomaly occurs in the routing protocol operations, such as a misconfiguration, a protocol loop, or an interface flapping issue, it can trigger a cascade of routing protocol updates, leading to the observed symptoms of route churn, adjacency loss, and LSP flooding. The PE node, being the central hub for routing protocol operations, is uniquely susceptible to these types of failures, making its prompt identification and resolution critical to maintaining network stability and ensuring high availability.

Pre-requisite Checks

- check: Verify the site power status and confirm that the Ericsson PE node HEB-PE-12 is receiving power.

- check: Check the status of all redundant links and ensure that they are operational.

- check: Confirm that the Hebbal Edge site is experiencing no other network-related issues.

Potential Root Causes

Cause	Probability
BGP Misconfiguration	High
OSPF Adjacency Loss due to Interface Flapping	Medium
ISIS LSP Flooding due to Routing Loop	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays a summary of the BGP sessions, including the number of established sessions, the number of routes received, and the number of routes advertised. This information can help identify potential issues with BGP sessions, such as failed sessions or routing information leaks.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays a list of OSPF neighbors, including their interface IP addresses, state, and timers.
```

```
This information can help identify potential issues with OSPF adjacency, such as failed adjacencies or timing issues.

Command: show isis database, Expected Output: ISIS database information, Description: This command displays a list of ISIS LSPs, including their LSP ID, state, and sequence number. This information can help identify potential issues with ISIS LSPs, such as flooding or corruption.
```

Resolution Steps

Step 1: Step: Verify BGP Configuration, Command: show bgp configuration, Description: Verify that the BGP configuration is correct and consistent with the expected settings. Check for any misconfigurations or inconsistencies that may be causing the routing protocol flap.

Step 2: Step: Reset OSPF Neighbor, Command: clear ospf neighbor, Description: Reset the OSPF neighbor relationship to force a re-establishment of the adjacency. This can help resolve issues related to OSPF adjacency loss due to interface flapping.

Step 3: Step: Clear ISIS LSP Database, Command: clear isis database, Description: Clear the ISIS LSP database to force a re-creation of LSPs. This can help resolve issues related to ISIS LSP flooding due to routing loops.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information, Description: Verify that the BGP sessions are stable and that the routing information is consistent.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that the OSPF adjacencies are stable and that the routing information is consistent.

Command: show isis database, Expected Output: ISIS database information, Description: Verify that the ISIS LSPs are stable and that the routing information is consistent.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: load /config/bgp.conf, Description: Restore the original BGP configuration from the configuration file /config/bgp.conf. This can help resolve issues related to BGP misconfiguration.

- Step: Re-establish OSPF Neighbor Relationship, Command: clear ospf neighbor, Description: Re-establish the OSPF neighbor relationship to force a re-establishment of the adjacency. This can help resolve issues related to OSPF adjacency loss due to interface flapping.

- Step: Restore Original ISIS LSP Database, Command: load /config/isis.db, Description: Restore the original ISIS LSP database from the configuration file /config/isis.db. This can help resolve issues related to ISIS LSP flooding due to routing loops.